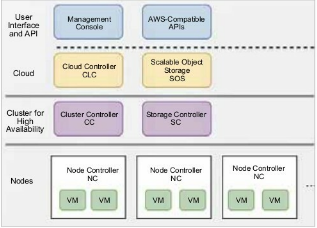

Figure 1: Eucalyptus software architecture

Networking: The Eucalyptus network is divided into three modes.

- *Managed mode:* In this mode, it just manages a local network of instances, which includes security groups and IP addresses.
- *System mode:* In this mode, it assigns a MAC address and attaches the instance's network interface to the physical network through the NC's bridge.
- *Static mode:* In this mode, it assigns IP addresses to instances.

Static and system mode do not assign elastic IPs, security groups, or VM isolation.

Access control is used to provide restriction to users. Each user will get a unique identity. All identities can be grouped and managed by access control.

Eucalyptus elastic block storage (EBS) provides block-level storage volumes, which we can attach to an instance.

Auto scaling and load balancing is used to automatically create or destroy instances or services based on requirements. CloudWatch provides different metrics for measurement.

Eucalyptus components

Eucalyptus has a total of six components, of which five are the main components and one is optional.

Cloud controller: Cloud controller (CLC) is the main controller, which manages the entire cloud platform. It provides a Web and EC2 compatible interface. All the incoming requests come through the Cloud controller. It performs scheduling, resource allocation and accounting. It manages all the underlying resources. Only one controller can exist per cloud.

Walrus: This is similar to AWS S3 (Simple Storage Service). It provides persistent storage to all the instances. It can contain any kind of data like application data, volume or image snapshots.

Cluster controller: This is the heart of the cluster within a Eucalyptus cloud. It manages VM (instance) execution and service level agreements. It communicates with the storage and network controller.

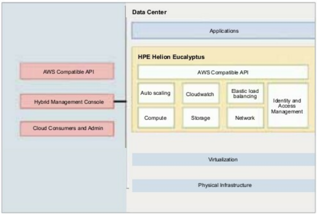

Figure 2: Eucalyptus components

Storage controller: Storage controller (SC) is similar to AWS EBS (Elastic Block Storage). It provides block level storage to instances and snapshots within a cluster. If an instance wants persistent storage outside of storage, then it must pass through Walrus. The storage controller doesn't handle this kind of request.

Node controller: NC (node controller) hosts all the instances and manages their end points. There is no limit to the number of NCs in the Eucalyptus cloud. It takes images and also caches from Walrus and creates instances. One should manage the number of NCs used as it affects the performance.

Enterprises can use any AWS-compatible tools or scripts to manage their own on-premise infrastructure. AWS API is implemented above Eucalyptus; so both are backward compatible. Users can run any apps that are supported by AWS from Eucalyptus.

Euca2ool: Euca2ool is the Eucalyptus CLI for interacting with Web services. It is a Python based tool which is compatible with all the AWS services like S3, auto scaling, ELB (Elastic Load Balancing), CloudWatch, EC2, etc. It is an all-in-one solution for both the AWS and the Eucalyptus platforms.

Other tools

There are many other tools that can be used to interact with Eucalyptus and AWS, and they are listed below.

s3curl*:* This is a tool for interaction between Eucalyptus Walrus and AWS S3.

Cloudberry S3 Explorer*:* This Windows tool is for managing files between Walrus and S3.

s3fs*:* This is a FUSE file system, which can be used to mount a bucket from S3 or Walrus as a local file system.

Vagrant AWS Plugin: This tool provides config files to manage AWS instances and also manage VMs on the local system.

You can refer to *https://github.com/eucalyptus/eucalyptus/wiki/AWS-tools* for more information.